

1) What do you mean by shear strength of soils and briefly explain how it is mobilized in soil mass

Ans) The shear strength of a soil is the resistance to deformation by continuous shear displacement of soil particles due to the action of shear stress. The shear strength is a most important characteristic of the soil.

The failure conditions for a soil may be expressed in terms of limiting shear stress called shear strength or as a function of the principal stresses

The shearing strength of soil is due to the following components

- a) **Structural resistance:** the structural arrangement of soil particles affects the shear stress because of the interlocking of the particles. The same soil may exhibit remarkably different shear strength at different void ratio and also at the different rate of loading
- b) **Frictional resistance:** the internal friction between soil particles resists the shearing friction of soil mass. The friction may be either sliding friction or a rolling friction or a combination of two
- c) **Cohesion :** The property of the soil holding its soil particles together is called cohesion. The soil which does not possess any cohesion is called cohesionless soil such as dry sand. The soil which possesses actual cohesion is called cohesion soil such as clays

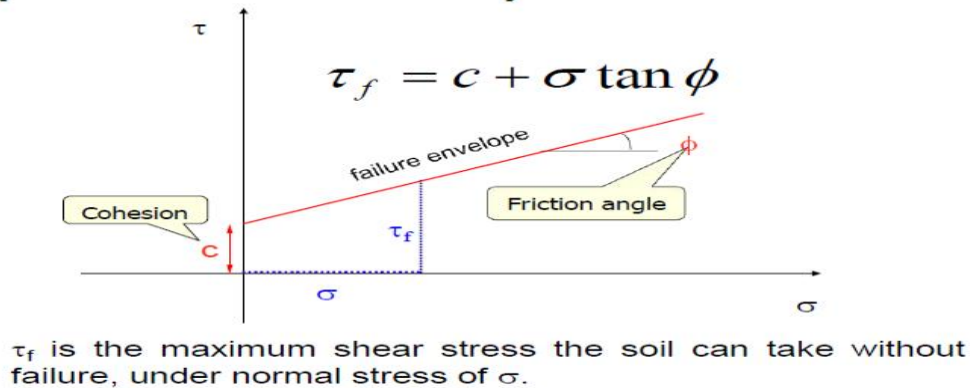
2) Describe major, intermediate and minor principal stress

Through a point in a loaded soil mass, innumerable planes pass and stress components on each plane depends upon the direction of the plane. It can be shown that there exists three typical planes, mutually orthogonal to each other, on which the stress is wholly normal and no shear stress acts. These planes are called the principal planes and the normal stresses acting on these planes are called the principal stresses.

In the order of decreasing magnitude of the normal stress, these planes are called major, intermediate and minor principal planes and the corresponding normal stresses on them are called major principal stress, intermediate principal stress and minor principal stress.

3) Explain Mohr Coulomb failure theory for shear strength of soil

Mohr-Coulomb Failure Criterion (in terms of **total stresses**)



The mohr coulomb theory is based on the following concepts

- i) Material fails essentially by shear. The critical shear stress causing failure depends upon the properties of the material as well as on normal stress on failure plane
- ii) The ultimate strength of the material is determined by the stresses on the potential failure plane
- iii) when the material is subjected to three dimensional principal stress. The intermediate principal stress does not have any influence on the strength of the material. In other words the failure criteria is independent of the intermediate principal stress.

This theory was first expressed by Coulomb and later generalized by Mohr. this theory can be expressed algebraically by the equation

$\tau_f = s$ = shear stress on failure plane of failure = shear resistance of material

$F(\sigma)$ = Function of normal stress

If the normal and shear stress corresponding to failure are plotted, then a curve is obtained. the plot or the curve is called the strength envelope. Coulomb defined the function $F(\sigma)$ as linear function of σ and gave the following strength equation

$$S = C + \sigma \tan \phi$$

4) Explain different factors affecting shear strength of cohesive soils

Solution) The factors affecting shear strength of cohesive soils are

- i) shape of particles: The shearing strength of sands with angular particles having sharp edges is greater than that with rounded particles, other parameters being identical
- ii) Gradation: A well graded sand exhibits greater shear strength than uniform sand
- iii) Confining pressure: The shear strength increases in confining pressure. However, for the range of pressures in the common field problems, the effect of confining pressure on the angle of shearing resistance is not significant.
- iv) Loading: The angle of shearing resistance of sand is independent of the rate of loading. The increase in the value of ϕ from the slowest to the fastest possible rate of loading is only about 1 to 2%
- v) Type of minerals: If the sand contains mica, it will have a large void ratio and a lower value of ϕ . However, it makes no difference whether the sand is composed of quartz or feldspar minerals
- vi) vibrations and repeated loading: Repeated loading can cause significant changes. A stress much smaller than the static failure stress if repeated a larger number of times can cause a very large strain and hence the failure.

5) Explain different factors affecting shear strength of cohesion less soils

Solution) The factors affecting shear strength of cohesion less soils are

- i) Structure of clay: The clay exhibits a definite structure. Even a normally consolidated clay exhibits a small peak due to structural strength. In case of over consolidated clays, the structural strength is predominant

5) Explain different factors affecting shear strength of cohesionless soils

Solution) The factors affecting shear strength of cohesionless soils are

- i) shape of particles: The shearing strength of sands with angular particles having sharp edges is greater than that with rounded particles, other particles being identical.
- ii) Gradation: well graded sand exhibits greater shear strength than a uniform sand
- iii) Density: Higher the density, higher will be shear strength due to low void ratio

iv) Confining pressure: The shear strength increases with an increase in confining pressure. However, for the range of pressures in the common field problems, the effect of confining pressure on the angle of shearing resistance is not significant.

v) Vibrations and repeated loading: Repeated loading can cause significant changes. A stress much smaller than the static failure stress if repeated a large number of times can cause a very large strain and hence the failure.

VI) type of minerals: If the sand contains mica, it will have large void ratio and a lower value of angle of internal friction. However, it makes no difference whether the sand is composed of quartz or feldspar minerals.

VI) Write short notes on

i) Direct shear test ii) Triaxial test iii) Vane shear test iv) Unconfined compressive test

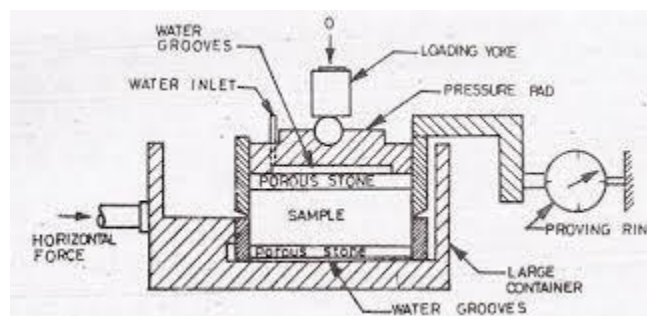
DIRECT SHEAR TEST

Objective

To determine the shearing strength of the soil using the direct shear apparatus.

Apparatus

Direct shear box apparatus, Loading frame, Dial gauge, Proving ring, Tamper, Balance to weigh up to 200 mg, Aluminum container, Spatula.



KNOWLEDGE OF EQUIPMENT:

Strain controlled direct shear machine consists of shear box, soil container, loading unit, proving ring, dial gauge to measure shear deformation and volume changes. A two piece square shear box is one type of soil container used.

A proving ring is used to indicate the shear load taken by the soil initiated in the shearing plane.

PROCEDURE

1. Check the inner dimension of the soil container.
2. Put the parts of the soil container together.
3. Calculate the volume of the container. Weigh the container.
4. Place the soil in smooth layers (approximately 10 mm thick). If a dense sample is desired tamp the soil.
5. Weigh the soil container, the difference of these two is the weight of the soil. Calculate the density of the soil.
6. Make the surface of the soil plane.
7. Put the upper grating on stone and loading block on top of soil.
8. Measure the thickness of soil specimen.
9. Apply the desired normal load.
10. Remove the shear pin.
11. Attach the dial gauge which measures the change of volume.
12. Record the initial reading of the dial gauge and calibration values.
13. Before proceeding to test check all adjustments to see that there is no connection between two parts except sand/soil.
14. Start the motor. Take the reading of the shear force and record the reading.
15. Take volume change readings till failure.
16. Add 5 kg normal stress 0.5 kg/cm^2 and continue the experiment till failure

17. Record carefully all the readings. Set the dial gauges zero, before starting the experiment

DATA CALCULATION SHEET FOR DIRECT SHEAR TEST

Normal stress 0.5 kg/cm^2 L.C=..... P.R.C=.....

Horizontal Gauge Reading (1)	Vertical Dial gauge Reading (2)	Proving ring Reading (3)	Hori.Dial gauge Reading Initial reading div. gauge (4)	Shear deformation Col.(4) x Leastcount of dial (5)	Vertical gauge reading Initial Reading (6)	Vertical deformation = div.in col.6 x L.C of dial gauge (7)	Proving ring reading Initial reading (8)	Shear stress = $\frac{\text{div.col.(8)} \times \text{proving ring constant Area of the specimen}(\text{kg/cm}^2)}{}$ (9)
0								
25								
50								
75								
100								
125								
150								
175								
200								
250								
300								
400								
500								
600								

700								
800								
900								

Normal stress 1.0 kg/cm^2 L.C=..... P.R.C=.....

TRIAXIAL TEST

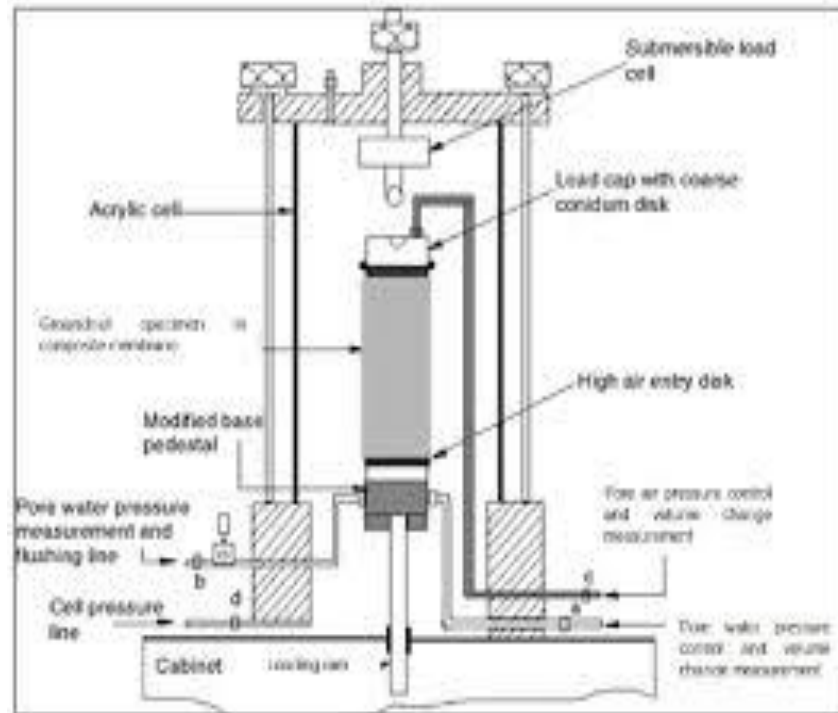
OBJECTIVE

To find the shear of the soil by Undrained Triaxial Test.

NEED AND SCOPE OF THE TEST

The standard consolidated undrained test is compression test, in which the soil specimen is first consolidated under all round pressure in the triaxial cell before failure is brought about by increasing the major principal stress.

It may be perform with or without measurement of pore pressure although for most applications the measurement of pore pressure is desirable.



PLANNING AND ORGANIZATION

Knowledge of Equipment

A constant rate of strain compression machine of which the following is a brief description of one is in common use.

- a) A loading frame in which the load is applied by a yoke acting through an elastic dynamometer, more commonly called a proving ring which used to measure the load. The frame is operated at a constant rate by a geared screw jack. It is preferable for the machine to be motor driven, by a small electric motor.
- b) A hydraulic pressure apparatus including an air compressor and water reservoir in which air under pressure acting on the water raises it to the required pressure, together with the necessary control valves and pressure dials.

A triaxial cell to take 3.8 cm dia and 7.6 cm long samples, in which the sample can be subjected to an all round hydrostatic pressure, together with a vertical compression load acting through a piston. The vertical load from the piston acts on a

pressure cap. The cell is usually designed with a non-ferrous metal top and base connected by tension rods and with walls formed of perspex.

Apparatus for preparation of the sample :

- a) 3.8 cm (1.5 inch) internal diameter 12.5 cm (5 inches) long sample tubes.
- b) Rubber ring.
- c) An open ended cylindrical section former, 3.8 cm inside dia, fitted with a small rubber tube in its side.
- d) Stop clock.
- e) Moisture content test apparatus.
- f) A balance of 250 gm capacity and accurate to 0.01 gm.

Experimental Procedure

1. The sample is placed in the compression machine and a pressure plate is placed on the top. Care must be taken to prevent any part of the machine or cell from jogging the sample while it is being setup, for example, by knocking against this bottom of the loading piston. The probable strength of the sample is estimated and a suitable proving ring selected and fitted to the machine.
2. The cell must be properly set up and uniformly clamped down to prevent leakage of pressure during the test, making sure first that the sample is properly sealed with its end caps and rings (rubber) in position and that the sealing rings for the cell are also correctly placed.
3. When the sample is setup water is admitted and the cell is fitted under water escapes from the bleed valve, at the top, which is closed. If the sample is to be tested at zero lateral pressure water is not required.

Cell pressure kg/cm ²	Strain dial 2	Proving ring reading	Load on sample kg	Corrected area cm ²	Deviator stress
1	2	3	4	5	6
0.5	0				

	50 100 150 200 250 300 350 400 450				
0.5	0 50 100 150 200 250 300 350 400 450				
0.5	0 50 100 150				

	200				
	250				
	300				
	350				
	400				
	450				

UNCONFINED COMPRESSION TEST

OBJECTIVE

determine shear parameters of cohesive soil

NEED AND SCOPE OF THE EXPERIMENT

It is not always possible to conduct the bearing capacity test in the field. Some times it is cheaper to take the undisturbed soil sample and test its strength in the laboratory. Also to choose the best material for the embankment, one has to conduct strength tests on the samples selected. Under these conditions it is easy to perform the unconfined compression test on undisturbed and remoulded soil sample. Now we will investigate experimentally the strength of a given soil sample.



PLANNING AND ORGANIZATION

We have to find out the diameter and length of the specimen.

EQUIPMENT

1. Loading frame of capacity of 2 t, with constant rate of movement. What is the least count of the dial gauge attached to the proving ring!
2. Proving ring of 0.01 kg sensitivity for soft soils; 0.05 kg for stiff soils.
3. Soil trimmer.
4. Frictionless end plates of 75 mm diameter (Perspex plate with silicon grease coating).
5. Evaporating dish (Aluminum container).
6. Soil sample of 75 mm length.
7. Dial gauge (0.01 mm accuracy).
8. Balance of capacity 200 g and sensitivity to weigh 0.01 g.
9. Oven, thermostatically controlled with interior of non-corroding material to maintain the temperature at the desired level. What is the range of the temperature used for drying the soil !
10. Sample extractor and split sampler.

11. Dial gauge (sensitivity 0.01mm).

12. Vernier calipers

EXPERIMENTAL PROCEDURE (SPECIMEN)

1. In this test, a cylinder of soil without lateral support is tested to failure in simple compression, at a constant rate of strain. The compressive load per unit area required to fail the specimen as called Unconfined compressive strength of the soil.

Preparation of specimen for testing

A. Undisturbed specimen

1. Note down the sample number, bore hole number and the depth at which the sample was taken.
2. Remove the protective cover (paraffin wax) from the sampling tube.
3. Place the sampling tube extractor and push the plunger till a small length of sample moves out.
4. Trim the projected sample using a wire saw.
5. Again push the plunger of the extractor till a 75 mm long sample comes out.
6. Cutout this sample carefully and hold it on the split sampler so that it does not fall.
7. Take about 10 to 15 g of soil from the tube for water content determination.
8. Note the container number and take the net weight of the sample and the container.
9. Measure the diameter at the top, middle, and the bottom of the sample and find the average and record the same.
10. Measure the length of the sample and record.

11. Find the weight of the sample and record.

B. Moulded sample

1. For the desired water content and the dry density, calculate the weight of the dry soil W_s required for preparing a specimen of 3.8 cm diameter and 7.5 cm long.

2. Add required quantity of water W_w to this soil.

$$W_w = W_s \times W/100 \text{ gm}$$

3. Mix the soil thoroughly with water.
4. Place the wet soil in a tight thick polythene bag in a humidity chamber and place the soil in a constant volume mould, having an internal height of 7.5 cm and internal diameter of 3.8 cm.
5. After 24 hours take the soil from the humidity chamber and place the soil in a constant volume mould, having an internal height of 7.5 cm and internal diameter of 3.8 cm.
6. Place the lubricated moulded with plungers in position in the load frame.
7. Apply the compressive load till the specimen is compacted to a height of 7.5 cm.
8. Eject the specimen from the constant volume mould.
9. Record the correct height, weight and diameter of the specimen.

Test procedure

1. Take two frictionless bearing plates of 75 mm diameter.
2. Place the specimen on the base plate of the load frame (sandwiched between the end plates).

3. Place a hardened steel ball on the bearing plate.
4. Adjust the center line of the specimen such that the proving ring and the steel ball are in the same line.
5. Fix a dial gauge to measure the vertical compression of the specimen.
6. Adjust the gear position on the load frame to give suitable vertical displacement.
7. Start applying the load and record the readings of the proving ring dial and compression dial for every 5 mm compression.
8. Continue loading till failure is complete.
9. Draw the sketch of the failure pattern in the specimen.

Project :

Tested by :

Location :

Boring No. :

Depth :

Sample details

Type UD/R : soil description

Specific gravity (G_s) 2.71

Bulk density

Water content

Degree of saturation .%

Diameter (D_o) of the sample cm

Area of cross-section = cm^2

Initial length (L_o) of the sample = 76 mm

Elapsed time (minutes) 1	Compression dial reading (L) (mm) 2	Strain L_0 (%) 3	Area $A_0 / (1 - e)^2$ (cm) 4	Proving ring reading (Divns.) 5	Axial load (kg) 6	Compressive stress (kg/cm ²) 7

Interpretation and Reporting

Unconfined compression strength of the soil = q_u =

Shear strength of the soil = $q_u/2$ =

Sensitivity = (q_u for undisturbed sample) / (q_u for remoulded sample).

Vane Shear Test

OBJECTIVE

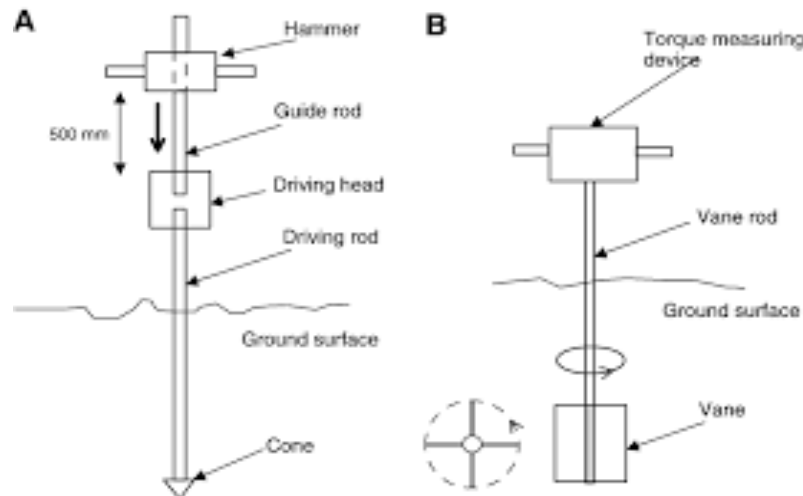
To find shear strength of a given soil specimen.

NEED AND SCOPE

The structural strength of soil is basically a problem of shear strength.

Vane shear test is a useful method of measuring the shear strength of clay. It is a cheaper and quicker method. The test can also be conducted in the laboratory. The

laboratory vane shear test for the measurement of shear strength of cohesive soils, is useful for soils of low shear strength (less than 0.3 kg/cm^2) for which triaxial or unconfined tests can not be performed. The test gives the undrained strength of the soil. The undisturbed and remoulded strength obtained are useful for evaluating the sensitivity of soil.



PLANNING AND ORGANIZATION

EQUIPMENT

- 1.Vane shear apparatus.
- 2.Specimen.
- 3.Specimen container.
- 4.Callipers.

EXPERIMENTAL PROCEDURE

- 1.Prepare two or three specimens of the soil sample of dimensions of at least 37.5 mm diameter and 75 mm length in specimen.(L/D ratio 2 or 3).
- 2.Mount the specimen container with the specimen on the base of the vane shear apparatus. If the specimen container is closed at one end, it should be provided with a hole of about 1 mm diameter at the bottom.
- 3.Gently lower the shear vanes into the specimen to their full length without disturbing the soil specimen. The top of the vanes should be atleast 10 mm below the top of the specimen. Note the readings of the angle of twist.

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